Algebra II with Trigonometry Chapter 11.3

Study guide

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1. Short Summary of the Document

This document is a section from an Algebra 2/Trigonometry Honors textbook on hyperbolas. It introduces the geometric and algebraic definitions of hyperbolas, describes their standard equations (both horizontal and vertical), and explains how to graph them by sketching the central box, asymptotes, vertices, and the curves themselves. The document provides properties of hyperbolas (such as vertices, foci, asymptotes, and axes) and guides students through finding equations and sketching graphs given certain conditions or features.

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2. Core Definitions, Theorems, and Formulas

Definitions

- **Hyperbola**: The set of all points in the plane where the *difference* of the distances to two fixed points (foci) is a constant.
- **Vertices**: Points where the hyperbola crosses its axis; for horizontal hyperbolas at $(\pm a, 0)$.
- **Transverse Axis**: Segment joining the two vertices, length $2a$.
- **Center**: The midpoint between the vertices (often the origin).
- **Branches**: The two separate parts of a hyperbola.
- **Asymptotes**: Diagonal lines that the hyperbola approaches but never crosses.

Formulas and Theorems

Standard Equations

- **Horizontal Hyperbola, Center at Origin:**

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

- **Vertical Hyperbola, Center at Origin:**

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

- **Relationship between a, b, and c:**

$$c^2 = a^2 + b^2 \quad \text{where } c \text{ is the distance from the center to each focus.}$$

- **Vertices:** $(\pm a, 0)$ for horizontal, $(0, \pm a)$ for vertical.
- **Foci:** $(\pm c, 0)$ for horizontal, $(0, \pm c)$ for vertical.
- **Asymptotes (horizontal):** $y = \pm \frac{b}{a} x$
- **Asymptotes (vertical):** $y = \pm \frac{a}{b} x$
- **Transverse axis length:** $2a$
- **No y-intercept for standard (horizontal) hyperbola ($x = 0$ yields no solution)**

Graphing Steps

1. Sketch the central box (rectangle at $(\pm a, \pm b)$). 2. Draw asymptotes as diagonals of the central box. 3. Plot the vertices. 4. Sketch the two branches of the hyperbola approaching asymptotes.

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3. Worked Study Guide

Step 1: Understanding the Geometric Definition

A hyperbola consists of all points where the absolute value of the *difference* of the distances to two fixed points (the foci) is a constant, $2a$:

$$|d(P, F_1) - d(P, F_2)| = 2a$$

Step 2: Identifying the Standard Equation

- If the foci are on the x-axis and center at the origin:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

- Swap x and y for a vertical transverse axis.

Step 3: Key Parameters

- a: Distance from center to vertex along the axis.
- b: Related to the rectangle and the asymptotes.
- c: Distance from center to focus, where $c^2 = a^2 + b^2$.

Step 4: Graphing a Hyperbola

Example: Graph $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

- $a^2 = 9 \implies a = 3$, $b^2 = 16 \implies b = 4$
- Center: (0, 0)
- Vertices: (±3, 0)
- Foci: $c^2 = 9 + 16 = 25 \implies c = 5$, so (±5, 0)
- Asymptotes: $y = \pm\frac{b}{a}x = \pm\frac{4}{3}x$
- Draw rectangle with corners (±3, ±4), draw diagonals, plot vertices, sketch branches approaching asymptotes.

Step 5: Determining the Equation Given Features

If you know vertices and foci/asymptotes:

- Find a from vertices, c from foci or b from slopes of asymptotes.
- Use the standard equation, plugging in values for a, b.

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4. Five Practice Problems & Answers

1. Find the center, vertices, foci, and asymptotes of the hyperbola $\frac{x^2}{25} - \frac{y^2}{9} = 1$.

Answer:

- Center: (0, 0)

- Vertices: (±5, 0)
- $c^2 = 25 + 9 = 34 \implies c = \sqrt{34}$, so Foci: (±$\sqrt{34}$, 0)
- Asymptotes: $y = \pm \frac{3}{5}x$

2. What is the standard form of the hyperbola with vertices at (±2, 0) and foci at (±5, 0)?

Answer:

- $a = 2$, $c = 5$, $b^2 = c^2 - a^2 = 25 - 4 = 21$
- Equation: $\frac{x^2}{4} - \frac{y^2}{21} = 1$

3. Write the equation of the hyperbola with center at (0,0), vertices at (0,±3) and asymptotes $y = \pm 2x$.

Answer:

- Vertices: vertical, so $\frac{y^2}{9} - \frac{x^2}{b^2} = 1$
- Asymptote: slope = 2 $\rightarrow \frac{a}{b} = 2 \rightarrow 3/b = 2 \rightarrow b = 1.5$
- Equation: $\frac{y^2}{9} - \frac{x^2}{2.25} = 1$

4. For the equation $\frac{y^2}{16} - \frac{x^2}{25} = 1$, identify the transverse axis and its length.

Answer:

- Vertical hyperbola, transverse axis is vertical.
- Length of transverse axis: $2a = 2 \times 4 = 8$

5. If the asymptotes of a hyperbola are $y = \pm 4x$ and the vertices are at (0, ±2), what is the equation?

Answer:

- Vertices: vertical, so $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$
- $a = 2$. Since asymptote slope is $a/b = 4 \implies 2/b = 4 \implies b = 0.5$
- Equation: $\frac{y^2}{4} - \frac{x^2}{0.25} = 1$

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5. Assumptions or Ambiguities

- While equations and symmetry are centered at the origin in examples, hyperbolas can be centered elsewhere, but those cases are not covered in this excerpt.
- Graphs in Example 2 are not described in detail, and so related practice problems use only text descriptions.
- The document assumes all hyperbolas are centered at the origin.
- The steps for non-standard forms (rotated, shifted) are not discussed here and are not included in practice problems.

If further detail or clarification is needed for non-origin centers, additional information would be required.